Aalok Kumar

Backend Software Engineer | Distributed Systems | AWS
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PROFESSIONAL SUMMARY

Backend Software Engineer with 3 years 4 months of experience building revenue-critical, compliance-heavy distributed systems across fintech and accounting SaaS. Shipped production AI document pipelines (AWS Textract), event-driven payment architectures (SQS), concurrent background job frameworks, multi-jurisdiction tax filing platforms, and addon-based billing infrastructure — serving 10,000+ organisations and 30,000+ users globally. Deep expertise in Java, Spring Boot, PostgreSQL, and AWS; consistent track record of measurable performance improvements in high-throughput, business-critical environments.

TECHNICAL SKILLS

Languages	Java, SQL, C++
Frameworks	Spring Boot, DGS (GraphQL), FastAPI, Spring Security, Hibernate, JPA, Maven, Gradle
AI / Data	AWS Textract, document intelligence pipelines, OCR, structured data ingestion, pgvector
Messaging / Async	AWS SQS, event-driven architecture, async job execution, thread-pool tuning, bulk processing
Databases	PostgreSQL, OracleDB, MySQL, MongoDB, Redis — query tuning, connection-pool optimisation
Cloud / DevOps	AWS (ECS, SQS, S3, CloudWatch, Textract), Docker, CI/CD, Linux, Git
Architecture	Microservices, distributed systems, RESTful APIs, MVC, OOP, Builder and Factory Patterns
Domain	Multi-jurisdiction tax filing, e-Sign, e-Invoicing, subscription and billing infrastructure

PROFESSIONAL EXPERIENCE

Backend Software Engineer — Tinvio Digital Services (Jaz Accounting, Singapore)

Remote | Apr 2025 – Present

- Architected **AI document intelligence pipeline** (AWS Textract) processing **1,000 docs/month/org** across 10,000+ organisations — automating invoice and bill extraction into structured PostgreSQL storage.
- Built **Magic Reconciliation engine** automatically matching bank statements against invoices, bills, and journal entries, eliminating manual reconciliation at scale.
- Engineered **background job framework** handling **1,000+ concurrent jobs**, crash-resilient with full state recovery across service restarts.
- Tuned thread-pool and async pipelines achieving **30% reduction in job processing time**; optimised transaction scopes maintaining **near-zero DB downtime** for 30,000+ users.
- Delivered **Philippines tax filing core**: 100+ daily filings, tax collections, and payment disbursements across **24 BIR tax form types**.
- Built **Singapore e-Invoicing** (50+ transactions/org/day) and **e-Sign core** with async document state machine and full audit trail.
- Designed **addon-based subscription and billing infrastructure** with plan enforcement, feature gating, and revenue-integrity guarantees via idempotent SQS payment event queues.
- Shipped 1,000+ pull requests spanning features, hotfixes, and refactors; led production incident investigations ensuring high availability.

Associate Consultant, Backend — Oracle Financial Services Software

Bengaluru, India | Jun 2023 – Mar 2025

- Customised core banking modules for **5+ enterprise banking clients**; optimised 20+ REST APIs and 100+ OracleDB stored procedures.
- Delivered solutions serving **100,000+ end users** on systems processing **1M+ transactions daily**.

EDUCATION

B.Tech, Electrical Engineering | CGPA: 8.2 / 10

National Institute of Technology, Hamirpur | Jul 2019 – May 2023

CERTIFICATIONS

- Oracle Certified Professional — Java SE Developer (Aug 2024)
- Oracle Certified Professional — Java EE Application Developer (Aug 2024)

ACHIEVEMENTS

- NCC Army 'B' Certificate (2021) and 'C' Certificate (2022) — leadership under structured command.
- 2nd place — 'Lost Lander' event: autonomous IoT bot solving a maze puzzle (EE + software integration).
- Navodaya Volunteer Teacher — mentored 25+ underprivileged students for JNV entrance examination.